CST 3526 Stochastic Process Lecture 6 - 10/29/2026

Lecture 6: Markov Processes II

Lecturer: Ziqiao Wang

In this lecture,[1] we first study stationary distributions and long-term behavior of discrete-time Markov chains, continuing from Lecture 5. We then allow state changes to occur at random times, introduce continuous-time Markov chains, and calculate their evolution and equilibrium probabilities through transition rates.

1 Stationary Distributions and Long-Term Behavior

We continue to use probability row vectors: $\mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}^n$.

1.1 Stationarity

Definition 1.1 (Stationary distribution)**.** A probability row vector $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots)$ is a *stationary distribution* if

$$\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}, \qquad \pi_j \geq 0, \qquad \sum_j \pi_j = 1. \tag{1}$$

The equation for component j is $\pi_j = \sum_i \pi_i p_{ij}$: the probability assigned to j equals the total probability entering j after one transition. The normalization condition is essential; the homogeneous linear equations alone also admit the zero vector.

Choosing $\mathbf{p}(0) = \boldsymbol{\pi}$ means drawing the initial state with probabilities $\pi_0, \pi_1, \ldots$. Then $\mathbf{p}(n) = \boldsymbol{\pi}\mathbf{P}^n = \boldsymbol{\pi}$ for every n. The state along an individual sample path can still change. For a homogeneous Markov chain, the path formula also shows that all finite-dimensional distributions are invariant under a common shift of time. This connects to stationary processes, which will be studied more fully in the later lectures on second-order processes.

Time homogeneity does not mean stationarity. Recall the speech model from Lecture 5, with state 0 for silence and state 1 for talking:

$$\mathbf{P} = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}. \tag{2}$$

It has the same transition matrix at every step. Yet, starting from silence, its marginal distributions change from $(1, 0)$ to $(3/4, 1/4)$ to $(11/16, 5/16)$. A fixed transition rule need not preserve an arbitrary initial distribution.

The stationary distribution of the urn chain. Recall the two-urn chain from Lecture 5, in state order $(0, 1, 2)$:

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1 & 0 \end{pmatrix}. \tag{3}$$

The first and third component equations give $\pi_0 = \pi_1/4$ and $\pi_2 = \pi_1/4$. Combining them with $\pi_0+\pi_1+\pi_2 = 1$ obtains

$$\boldsymbol{\pi} = (1/6,\ 2/3,\ 1/6).$$

[1] Main reading: Leon-Garcia, Sections 11.2.3, 11.3.3, and 11.4.

The middle equation $\pi_1 = \pi_0 + \pi_1/2 + \pi_2$ is then satisfied. The three possible counts are not equally likely in equilibrium. **The weights match the $\binom{4}{2} = 6$ labeled configurations: one has no white balls in urn 1, four have one, and one has two.**

1.2 Stationary and Limiting

A *stationary distribution* is preserved if the chain starts with that distribution. A *limiting distribution* is what $\mathbf{p}(n) = \mathbf{p}(0)\mathbf{P}^n$ approaches from a *specified initial distribution*, if the limit exists. Solving $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ identifies a distribution that stays fixed; it does not show that other initial distributions approach it.

Theorem 1.1 (Finite irreducible chains)**.** *A finite irreducible Markov chain has a unique stationary distribution $\boldsymbol{\pi}$, and every π_j is positive. If the chain is also aperiodic, then*

$$\lim_{n\to\infty} p_{ij}(n) = \pi_j \quad \text{for every } i, j, \qquad \lim_{n\to\infty} \mathbf{p}(0)\mathbf{P}^n = \boldsymbol{\pi} \tag{4}$$

for every initial distribution $\mathbf{p}(0)$.

The theorem has two conclusions: **finite irreducibility guarantees *existence and uniqueness*; adding aperiodicity guarantees *convergence from every initial distribution.*** We omit the general proof and later verify convergence directly for the two-state speech model.

The urn chain is finite, irreducible, and aperiodic. Its distribution therefore tends to $(1/6, 2/3, 1/6)$ from any initial count. The following two examples explain why the theorem's assumptions matter.

Example 1. For the alternating chain from Lecture 5,

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad (a, 1-a)\mathbf{P} = (1-a, a). \tag{5}$$

This equals $(a, 1-a)$ only when $a = 1/2$, so the unique stationary distribution is $(1/2, 1/2)$. However, starting with $X_0 = 0$ gives

$$\mathbf{p}(0) = (1, 0), \quad \mathbf{p}(1) = (0, 1), \quad \mathbf{p}(2) = (1, 0), \quad \mathbf{p}(3) = (0, 1), \quad \ldots$$

The distribution keeps alternating and does not approach $(1/2, 1/2)$. Thus there is no limiting distribution *for this initial condition.*

Starting instead with $\mathbf{p}(0) = (1/2, 1/2)$ means choosing each starting state with probability $1/2$. The only possible sample paths are

probability	X_0	X_1	X_2	X_3	$\cdots$
1/2	0	1	0	1	$\cdots$
1/2	1	0	1	0	$\cdots$

In every column, the probabilities of 0 and 1 are each $1/2$. Therefore $\mathbf{p}(n) = (1/2, 1/2)$ for every n, even though each sample path changes at every step. **A stationary distribution does not mean a constant sample path.** This chain is irreducible but has period two, so the theorem's aperiodicity assumption fails.

Example 2. Recall the absorbing chain, in state order $(0, 1, 2)$:

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 0 & 1 \end{pmatrix}. \tag{6}$$

States 0 and 2 never change. Consequently,

$$(q, 0, 1-q)\mathbf{P} = (q, 0, 1-q), \qquad 0 \le q \le 1.$$

Each of these vectors is a stationary distribution: choose state 0 initially with probability q and state 2 with probability $1-q$, and that distribution stays fixed forever. In fact, the stationary equation for state 1 gives $\pi_1 = \pi_1/2$, hence $\pi_1 = 0$; these are all the stationary distributions.

Now start from $X_0 = 1$, so $\mathbf{p}(0) = (0, 1, 0)$. To be in state 1 after n steps, the chain must take its self-loop at every step; once it leaves, it cannot come back. Thus $\mathbb{P}(X_n = 1 \mid X_0 = 1) = 2^{-n}$. At every possible exit step, the probabilities of going to 0 and 2 are equal. The remaining probability $1 - 2^{-n}$ is therefore split equally between them:

$$\mathbf{p}(n) = \left(\frac{1 - 2^{-n}}{2},\ 2^{-n},\ \frac{1 - 2^{-n}}{2} \right) \longrightarrow (1/2, 0, 1/2).$$

By contrast, starting from 0 gives $\mathbf{p}(n) = (1, 0, 0)$ for every n, and starting from 2 gives $(0, 0, 1)$. Here limits exist, but they depend on the initial state. This reducible chain has multiple stationary distributions; reducibility *can*, but need not, cause nonuniqueness.

1.3 Solving the two-state model completely

Return to the general two-state speech model,

$$\mathbf{P} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix}, \qquad 0 < \alpha, \beta < 1. \tag{7}$$

The stationary equations reduce to

$$\alpha\pi_0 = \beta\pi_1, \qquad \pi_0 + \pi_1 = 1,$$

so

$$\boldsymbol{\pi} = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta} \right). \tag{8}$$

To find the whole evolution rather than just a candidate limit, put $u_n = \mathbb{P}(X_n = 1)$. Conditioning on X_n gives

$$u_{n+1} = \alpha(1 - u_n) + (1 - \beta)u_n = \alpha + (1 - \alpha - \beta)u_n.$$

The fixed point is $u_* = \alpha/(\alpha + \beta)$. Subtracting it from the recurrence,

$$u_{n+1} - u_* = (1 - \alpha - \beta)(u_n - u_*).$$

Iterating gives

$$u_n = \frac{\alpha}{\alpha+\beta} + (1 - \alpha - \beta)^n \left(u_0 - \frac{\alpha}{\alpha+\beta} \right). \tag{9}$$

Since $|1 - \alpha - \beta| < 1$, the initial-condition term tends to zero. The limiting speech probability is therefore $\pi_1 = \alpha/(\alpha + \beta)$, independently of u_0.

For $\alpha = 1/4$, $\beta = 1/2$, starting from silence means $u_0 = 0$, and

$$\mathbb{P}(X_n = 1) = \frac{1}{3}\left[1 - \left(\frac{1}{4} \right)^n \right] \longrightarrow \frac{1}{3}.$$

At $n = 1, 2$, this gives $1/4$ and $5/16$, agreeing with the earlier matrix calculations. The limiting distribution is $(2/3, 1/3)$.

Q: If this chain starts in its stationary distribution, are successive states independent?

A: No. The marginal speech probability is always $1/3$, but

$$\mathbb{P}(X_{n+1} = 1 \mid X_n = 1) = \frac{1}{2} \neq \frac{1}{3}.$$

Stationarity preserves distributions over time; it does not remove temporal dependence.

2 From Discrete Steps to Continuous Time

The speech model changes state at integer time slots. A machine can fail or finish a repair at any real time. To describe such a process, we must answer two questions: *How long does the current state last? Where does the next jump go?*

Definition 2.1 (Continuous-time Markov chain)**.** A process $\{X(t) : t \geq 0\}$ with a finite or countable state space S is a continuous-time Markov chain (CTMC) if, for $0 \leq t_0 < \cdots < t_k < t$,

$$\mathbb{P}(X(t) = j \mid X(t_k) = i_k, \ldots, X(t_0) = i_0) = \mathbb{P}(X(t) = j \mid X(t_k) = i_k)$$

whenever the conditioning history has positive probability. It is *time-homogeneous* if

$$p_{ij}(t) = \mathbb{P}(X(s+t) = j \mid X(s) = i)$$

does not depend on the starting time s. Here t is an elapsed duration, not a number of jumps.

For the rest of this lecture, S is finite. A sample path is constant between jumps; at a jump time it takes the new state value (*right-continuity*). Continuous time does not mean that the sample path is a continuous function.

2.1 State occupancy times

Start in state i at time 0. Let

$$H_i = \inf\{t > 0 : X(t) \neq i\}$$

be the time until the *first departure* from i, with $H_i = \infty$ if there is no departure. This is the *state occupancy time*, also called the *holding time*. In this subsection, probabilities are conditional on $X(0) = i$.

Define its *survival function*, or tail probability, by

$$G_i(t) = \mathbb{P}(H_i > t) = 1 - F_{H_i}(t), \qquad t \geq 0.$$

It is the probability that this uninterrupted visit has not ended by time t. In particular, $G_i(0) = 1$.

To wait longer than $s + t$, we must first wait longer than s, then wait longer than another t:

$$\mathbb{P}(H_i > s + t) = \mathbb{P}(H_i > s)\mathbb{P}(H_i > s + t \mid H_i > s).$$

On $\{H_i > s\}$ the current state is still i. The Markov property removes dependence on the past visit, and time homogeneity makes a fresh wait at time s have the same law as at time 0. Thus

$$\mathbb{P}(H_i > s + t \mid H_i > s) = G_i(t), \qquad G_i(s + t) = G_i(s)G_i(t). \tag{10}$$

This is *memorylessness*: how long we have already waited does not change the distribution of the remaining wait.

Put $a = G_i(1)$. We have $a > 0$[2]. Repeated use of Eq. (10) now gives

$$G_i(1/m) = a^{1/m}, \qquad G_i(n/m) = a^{n/m}.$$

Since a tail probability is nonincreasing, approximation from above and below by rational numbers extends this identity to every real $t \geq 0$. Define $v_i = -\ln a \geq 0$, where one unit of time is used in $G_i(1)$. Then

$$G_i(t) = e^{-v_i t}, \qquad H_i \sim \text{Exp}(v_i), \qquad \mathbb{E}[H_i] = \frac{1}{v_i} \quad (v_i > 0). \tag{11}$$

[2]Otherwise $G_i(1) = G_i(1/m)^m = 0$ would give $G_i(1/m) = 0$ for every integer m, contradicting $G_i(h) \to 1$ as $h \to 0$.

The density is $v_i e^{-v_i t}$, and integrating t times this density gives the mean. Thus v_i is the inverse mean occupancy time. It has units of inverse time. If $v_i = 0$, then $G_i(t) = 1$ for every finite t: the state is absorbing and $H_i = \infty$.

Keep v_i fixed and let $h \to 0$. By Taylor's theorem, retaining the constant and linear terms, gives

$$G_i(h) = e^{-v_i h} = 1 - v_i h + r_i(h), \qquad \text{where } r_i(h) = \frac{v_i^2}{2} e^{-v_i \xi_h} h^2, \quad 0 < \xi_h < h.$$

The point ξ_h may depend on h; we do not need to find it. Because $v_i \geq 0$,

$$0 \leq \frac{r_i(h)}{h} = \frac{v_i^2}{2} e^{-v_i \xi_h} h \leq \frac{v_i^2}{2} h \longrightarrow 0.$$

Thus the error left after the linear approximation becomes negligible compared with h itself. This is what the notation *little-o* expresses, that is

$$r(h) = o(h) \quad \text{as } h \to 0 \quad \Longleftrightarrow \quad \lim_{h \to 0} \frac{r(h)}{h} = 0.$$

It is not enough that $r(h) \to 0$: for example, $h^2 = o(h)$ because $h^2/h = h \to 0$, whereas h is not $o(h)$ because $h/h = 1$. The symbol $o(h)$ denotes a remainder with this limiting property, not one fixed function; different occurrences can represent different remainders. We can therefore write the exponential tail

$$G_i(h) = 1 - v_i h + o(h), \qquad \text{and } 1 - G_i(h) = v_i h + o(h).$$

Setting $s = h$ in Eq. (10), subtracting $G_i(t)$ and dividing by h, we obtain

$$\frac{G_i(t+h) - G_i(t)}{h} = G_i(t) \frac{G_i(h) - 1}{h} \longrightarrow -v_i G_i(t).$$

This gives $G_i'(t) = -v_i G_i(t)$.

2.2 Embedded jump chain

For $v_i > 0$, let J_i be the state reached at the first departure from i. Define

$$\widetilde{q}_{ij} = \mathbb{P}(J_i = j \mid X(0) = i), \qquad \widetilde{q}_{ii} = 0, \qquad \sum_{j \neq i} \widetilde{q}_{ij} = 1. \tag{12}$$

These numbers describe *where the next actual jump goes.*

Given $X(0) = i$, the holding time and next destination are independent. Indeed, conditional on $H_i > s$, the process is still at i and its future starts afresh. Therefore

$$\mathbb{P}(H_i > s, J_i = j \mid X(0) = i) = G_i(s) \widetilde{q}_{ij}.$$

This factorization, for every $s \geq 0$, gives the stated independence. It is conditional on the current state; different states can have different waiting-time and destination laws.

Now record only the successive states at actual jumps. If T_n is the time of the nth change of state and $T_0 = 0$, set $Y_n = X(T_n)$. For example,

time interval	$[0, 0.2)$	$[0.2, 1.5)$	$[1.5, 2.1)$	$[2.1, \ldots)$
$X(t)$	0	2	1	0

$$\Longrightarrow \quad (Y_0, Y_1, Y_2, Y_3) = (0, 2, 1, 0).$$

The sequence $\{Y_n\}$ is the *embedded jump chain*: its index counts jumps, not units of time. Restarting the Markov process at each jump gives $\mathbb{P}(Y_{n+1} = j \mid Y_n = i, \text{ earlier states}) = \widetilde{q}_{ij}$. It is generally different from the fixed-time sample $X(0), X(1), X(2), \ldots$.

A CTMC can therefore be generated by two operations at state i: wait an exponential time of rate v_i, then independently choose the next state using $\widetilde{q}_{ij}$. Repeat after each jump. At an absorbing state, actual jumps stop; no next destination is needed.

2.3 From occupancy times to transition rates

Write $\mathbf{P}(t) = [p_{ij}(t)]$. Each row is a conditional probability distribution, so

$$\sum_j p_{ij}(t) = 1, \qquad p_{ij}(t) \geq 0, \qquad p_{ii}(0) = 1, \quad p_{ij}(0) = 0 \ (j \neq i).$$

In particular, $\mathbf{P}(0) = \mathbf{I}$.

For a nonabsorbing state i and $j \neq i$, the preceding independence gives the exact identity

$$\mathbb{P}(H_i \leq h, J_i = j \mid X(0) = i) = (1 - e^{-v_i h})\widetilde{q}_{ij}.$$

This is not generally $p_{ij}(h)$: after the first jump, another jump may occur before time h. However, this distinction is of smaller order for small h. To see why, put $M = \max_i v_i < \infty$. The first jump has probability at most Mh; after that jump, the conditional probability of another jump in the remaining time is also at most Mh. Hence

$$\mathbb{P}(\text{at least two jumps in } [0, h] \mid X(0) = i) \leq M^2 h^2 = o(h).$$

Using $1 - e^{-v_i h} = v_i h + o(h)$, we obtain

$$p_{ij}(h) = v_i \widetilde{q}_{ij} h + o(h), \qquad \gamma_{ij} := v_i \widetilde{q}_{ij} = \lim_{h \to 0} \frac{p_{ij}(h)}{h} \quad (j \neq i). \tag{13}$$

Thus the *transition rate* γ_{ij} combines the speed of leaving with the probability of choosing that destination. For an absorbing state all outgoing rates are zero. For a nonabsorbing state,

$$v_i = \sum_{j \neq i} \gamma_{ij}, \qquad \widetilde{q}_{ij} = \frac{\gamma_{ij}}{v_i} \quad (j \neq i). \tag{14}$$

The denominator normalizes rates into probabilities.

Note that probability conservation determines the diagonal expansion:

$$p_{ii}(h) = 1 - \sum_{j \neq i} p_{ij}(h) = 1 - v_i h + o(h), \qquad \gamma_{ii} := \lim_{h \to 0} \frac{p_{ii}(h) - 1}{h} = -v_i. \tag{15}$$

The initial value is 1 on the diagonal, so we subtract 1 before taking the derivative. The absolute value $-\gamma_{ii}$ is the total leaving rate; γ_{ii} is not a negative probability or a rate of actual self-jumps.

Definition 2.2 (Generator or transition-rate matrix)**.** The matrix $\mathbf{\Gamma} = [\gamma_{ij}]$ is the *generator*. Its off-diagonal entries are nonnegative, and

$$\sum_j \gamma_{ij} = -v_i + \sum_{j \neq i} \gamma_{ij} = 0, \qquad \mathbf{P}(h) = \mathbf{I} + h\mathbf{\Gamma} + o(h). \tag{16}$$

The matrix expansion collects the scalar expansions: diagonal entries start at 1, all other entries at 0. Equivalently, $\mathbf{\Gamma} = \lim_{h \to 0} [\mathbf{P}(h) - \mathbf{I}]/h$.

For example, a generator row $(-5, 2, 3)$, in state order $(0, 1, 2)$ and with time in minutes, means

$$\mathbb{E}[H_0] = 1/5 \text{ minute}, \qquad \widetilde{q}_{01} = 2/5, \quad \widetilde{q}_{02} = 3/5.$$

For $h = 0.01$ minute, the corresponding probability row is approximately $(0.95, 0.02, 0.03)$. The rate $\gamma_{01} = 2$ has units minute^{-1}; $2h$ is dimensionless and approximates a probability only for small h. This does not mean that a one-minute transition has probability 2, or that every minute contains exactly two such jumps.

For any duration t, the no-departure probability and the endpoint probability satisfy

$$\mathbb{P}(H_i > t \mid X(0) = i) = e^{-v_i t} \leq p_{ii}(t).$$

The right side also counts paths that leave i and return. Likewise, $1 - e^{-v_i t}$ is a first-departure probability, not generally the probability of being in another specified state at time t.

3 Evolution Equations and Equilibrium

3.1 Updating a distribution over a short time

Let $p_j(t) = \mathbb{P}(X(t) = j)$ and let $\mathbf{p}(t)$ be the probability row vector. Unlike $p_{ij}(t)$, this probability averages over the initial state. The law of total probability gives

$$p_j(t) = \sum_i p_i(0)p_{ij}(t), \qquad \mathbf{p}(t) = \mathbf{p}(0)\mathbf{P}(t).$$

At time t, the same argument over the next interval of length h gives

$$\begin{aligned} p_j(t+h) &= \sum_i p_i(t)p_{ij}(h) \\ &= p_j(t)(1 - v_j h) + \sum_{i \neq j} p_i(t)\gamma_{ij} h + o(h). \end{aligned}$$

Subtract $p_j(t)$, divide by h, and let $h \to 0$. Then

$$p_j'(t) = \underbrace{\sum_{i \neq j} p_i(t)\gamma_{ij}}_{\text{probability inflow rate}} - \underbrace{p_j(t)v_j}_{\text{probability outflow rate}}. \tag{17}$$

The factor $p_i(t)$ is essential: a transition out of i can contribute only when the process is currently in i. For example, $p_i(t)\gamma_{ij}h$ is the first-order probability of being at i at time t and moving to j during the short interval.

Using $\gamma_{jj} = -v_j$, Eq. (17) is the jth component of

$$\mathbf{p}(t) = \mathbf{p}(0)\mathbf{P}(t), \qquad \mathbf{p}'(t) = \mathbf{p}(t)\mathbf{\Gamma}. \tag{18}$$

A differential equation here describes the *rate of change of probabilities*.

3.2 Conditional probabilities: forward and backward equations

To find a transition over $s + t$, condition on the state at the intermediate time s:

$$p_{ij}(s+t) = \sum_k p_{ik}(s)p_{kj}(t), \qquad \mathbf{P}(s+t) = \mathbf{P}(s)\mathbf{P}(t). \tag{19}$$

The intermediate time is fixed; the number of jumps before it is random. Put the small interval at the end, or at the beginning, respectively:

$$\begin{aligned} \frac{\mathbf{P}(t+h) - \mathbf{P}(t)}{h} &= \mathbf{P}(t)\frac{\mathbf{P}(h) - \mathbf{I}}{h} \longrightarrow \mathbf{P}(t)\mathbf{\Gamma}, \\ \frac{\mathbf{P}(h+t) - \mathbf{P}(t)}{h} &= \frac{\mathbf{P}(h) - \mathbf{I}}{h}\mathbf{P}(t) \longrightarrow \mathbf{\Gamma}\mathbf{P}(t). \end{aligned}$$

We have therefore derived the *Kolmogorov forward and backward equations*:

$$\underbrace{\mathbf{P}'(t) = \mathbf{P}(t)\mathbf{\Gamma}}_{\text{forward}}, \qquad \underbrace{\mathbf{P}'(t) = \mathbf{\Gamma}\mathbf{P}(t)}_{\text{backward}}, \qquad \mathbf{P}(0) = \mathbf{I}. \tag{20}$$

"Backward" refers to conditioning near the beginning of the interval, not to running the sample path backward in time.

3.3 Stationary probabilities and probability flow

A probability row vector $\boldsymbol{\pi}$ is *stationary* if choosing $\mathbf{p}(0) = \boldsymbol{\pi}$ gives $\mathbf{p}(t) = \boldsymbol{\pi}$ for every $t \geq 0$, that is, $\boldsymbol{\pi}\mathbf{P}(t) = \boldsymbol{\pi}$. For our finite-state model this is equivalent to

$$\boldsymbol{\pi}\boldsymbol{\Gamma} = \mathbf{0}, \qquad \pi_j \geq 0, \qquad \sum_j \pi_j = 1. \tag{21}$$

Necessity. A fixed distribution has zero derivative. More explicitly,

$$\boldsymbol{\pi}\boldsymbol{\Gamma} = \lim_{h \to 0} \frac{\boldsymbol{\pi}\mathbf{P}(h) - \boldsymbol{\pi}}{h} = \mathbf{0}.$$

Sufficiency. Suppose $\boldsymbol{\pi}\boldsymbol{\Gamma} = \mathbf{0}$. The *backward* equation gives

$$\frac{d}{dt}[\boldsymbol{\pi}\mathbf{P}(t)] = \boldsymbol{\pi}\boldsymbol{\Gamma}\mathbf{P}(t) = \mathbf{0}.$$

Each component has zero derivative, so it is constant. Its value at $t = 0$ is $\boldsymbol{\pi}\mathbf{I} = \boldsymbol{\pi}$, proving $\boldsymbol{\pi}\mathbf{P}(t) = \boldsymbol{\pi}$.

Writing the jth component of $\boldsymbol{\pi}\boldsymbol{\Gamma} = \mathbf{0}$ and moving the diagonal term to the other side gives

$$0 = \sum_{i \neq j} \pi_i \gamma_{ij} - \pi_j v_j, \qquad \sum_{i \neq j} \pi_i \gamma_{ij} = \pi_j v_j. \tag{22}$$

These are the *global balance equations*: total probability inflow to each state equals total outflow. They do not require each pair of states to balance separately. For example, in the directed cycle $0 \to 1 \to 2 \to 0$ with all three rates equal to 1, the uniform distribution is stationary. Each state receives and sends probability at rate $1/3$, although there is no reverse transition on any edge. The stronger pairwise condition $\pi_i \gamma_{ij} = \pi_j \gamma_{ji}$ is called *detailed balance*; it fails in this example.

Theorem 3.1 (Finite irreducible CTMC)**.** *For a finite-state CTMC with finite rates, if every state is reachable from every other along positive-rate transitions, there is a unique stationary distribution $\boldsymbol{\pi}$. Moreover,*

$$\lim_{t \to \infty} p_{ij}(t) = \pi_j$$

for every pair i, j, and $\mathbf{p}(t) \to \boldsymbol{\pi}$ from every initial distribution.

To understand the absence of an extra periodicity condition, fix a time spacing $h > 0$. The *sampled chain* $X(0), X(h), X(2h), \ldots$ has transition matrix $\mathbf{P}(h)$.

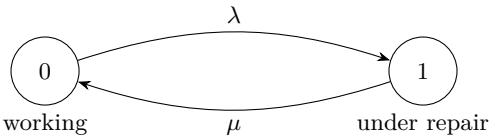
First, this sampled chain has a self-loop at every state, because

$$p_{ii}(h) \geq \mathbb{P}(H_i > h \mid X(0) = i) = e^{-v_i h} > 0.$$

Thus it is aperiodic. Second, every positive-rate path from i to j can be completed within h with positive probability: its finitely many exponential waits can all be short, and after arrival the process can stay at j for the remaining time. Hence the sampled chain is also irreducible. Its aperiodicity is therefore automatic, allowing the discrete-time convergence theorem to be used. In particular, alternating jump destinations $0, 1, 0, 1, \ldots$ do not force alternation at fixed observation times, since the number of jumps between observations is random.

4 A Two-State Failure–Repair Model

Example 3 (A machine that fails and is repaired)**.** Let $X(t) = 0$ mean working and $X(t) = 1$ mean under repair. Assume successive working periods are exponential with rate $\lambda > 0$, successive repair periods are exponential with rate $\mu > 0$, and all these durations are independent. Memorylessness means that the remaining working or repair time depends only on the present status, so the status is a sufficient Markov state.



The only destination from 0 is 1, and the only destination from 1 is 0. Thus $v_0 = \lambda$, $v_1 = \mu$, and

$$\mathbf{\Gamma} = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}. \tag{23}$$

Put $u(t) = \mathbb{P}(X(t) = 1)$, so the working probability is $1 - u(t)$. To be under repair at $t + h$, the machine can be working at t and fail during the short interval, or be under repair at t and still be under repair at its end. Therefore

$$u(t + h) = [1 - u(t)]\lambda h + u(t)(1 - \mu h) + o(h).$$

The $o(h)$ term includes paths with multiple changes in the interval. Subtracting $u(t)$, dividing by h, and letting $h \to 0$ gives

$$u'(t) = \lambda[1 - u(t)] - \mu u(t) = \lambda - (\lambda + \mu)u(t).$$

The two terms are the probability flow into and out of the repair state.

Set $a = \lambda + \mu$ and $u_* = \lambda/a$. The preceding equation becomes $u'(t) = -a[u(t) - u_*]$. Multiplying the deviation from equilibrium by e^{at} gives

$$\begin{aligned} \frac{d}{dt}\left(e^{at}[u(t) - u_*]\right) &= ae^{at}[u(t) - u_*] + e^{at}u'(t) \\ &= e^{at}\big(a[u(t) - u_*] - a[u(t) - u_*]\big) = 0. \end{aligned}$$

A function with zero derivative is constant. Evaluating that constant at $t = 0$, $e^{at}[u(t) - u_*] = u(0) - u_*$. Hence

$$u(t) = \frac{\lambda}{\lambda + \mu} + \left[u(0) - \frac{\lambda}{\lambda + \mu}\right] e^{-(\lambda+\mu)t}. \tag{24}$$

Starting from working means $u(0) = 0$, so $u(t) = p_{01}(t)$. Starting from repair means $u(0) = 1$, so $u(t) = p_{11}(t)$. Substituting these two initial conditions gives

$$\begin{aligned} p_{01}(t) &= \frac{\lambda}{a}(1 - e^{-at}), \quad p_{00}(t) = 1 - p_{01}(t), \\ p_{11}(t) &= \frac{\lambda}{a} + \frac{\mu}{a}e^{-at}, \quad p_{10}(t) = 1 - p_{11}(t) = \frac{\mu}{a}(1 - e^{-at}). \end{aligned}$$

Since $a > 0$, $e^{-at} \to 0$. The limiting distribution is independent of the initial state. It can also be found without calculating $u(t)$: balance the two stationary flows and normalize,

$$\lambda\pi_0 = \mu\pi_1, \qquad \pi_0 + \pi_1 = 1, \qquad \boldsymbol{\pi} = \left(\frac{\mu}{\lambda + \mu}, \frac{\lambda}{\lambda + \mu}\right). \tag{25}$$

For $\lambda = 1$ per hour and $\mu = 3$ per hour, the mean working period is one hour and the mean repair period is $1/3$ hour. Thus $\boldsymbol{\pi} = (3/4, 1/4)$, and a machine initially working is under repair at time t hours with probability $(1 - e^{-4t})/4$.

Its states alternate at every jump, so its transition matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and its stationary distribution is $(1/2, 1/2)$. It counts successive visits equally. The CTMC stationary law, computed above, reflects the time spent in each state. For $\lambda = 1$ and $\mu = 3$, a working visit lasts three times as long on average as a repair visit, consistent with the CTMC probabilities $(3/4, 1/4)$. The jump chain has period two, but the explicit formula for $u(t)$ shows that the CTMC distribution converges.